

PDA restrictive vs unrestrictive

Summary: A restrictive ductus arteriosus is defined hemodynamically as a vessel that imposes enough intrinsic resistance to maintain a discernible pressure gradient between the systemic and pulmonary circulations. Usually, when the duct is "unrestrictive", it allows free passage of flow and pressure equalization between the 2 compartments: by definition, the pressure in the pulmonary artery and the pressure in the aorta will equalize throughout the cardiac cycle since the 2 circuits are connected and the direction of the flow via the large unrestrictive ductus is largely dictated by the PVR/SVR ratio and/or filling pressure in these vessels - ex: interrupted aortic arch). The intrinsic ductal resistance is primarily determined by the internal diameter and the length of the ductal lumen. Unlike a large unrestrictive duct, which allows for the equalization of pressures between the aorta and the pulmonary artery, a restrictive duct prevents such equalization to various degree. Of course, this "restriction" should be viewed as a spectrum rather than a fixed anatomical property. Functional restriction can arise also when very high flow is driven through the ductus, generating shear stress, flow acceleration, and energy loss along its course. In this context, resistance is not solely determined by ductal size, but also by the magnitude and velocity of flow. A useful analogy is attempting to channel a massive volume of water—like Niagara Falls—through a conduit: even with a large diameter for the expected structure, sufficiently high flow will generate turbulence and resistance. Clinically, this can be seen in conditions such as large arteriovenous malformations, where markedly increased pulmonary blood flow overwhelms the ductal capacity, resulting in accelerated, high-velocity flow across the PDA despite a relatively large caliber. Flow through a restrictive duct is driven by the pressure difference between the two vascular beds rather than just the ratio of their resistances. Echocardiographic proof of restriction is typically established through Doppler interrogation of blood flow velocity - although there are some caveats. A restrictive duct is usually characterized by acceleration and a "high"-velocity jet, often exceeding 2 meters per second in one of the directions. According to the modified Bernoulli equation, these high velocities are reflective of a significant pressure drop across the narrow orifice. A small ductus with intrinsically high resistance does not necessarily generate high Doppler velocities if the driving pressure gradient is limited. In low cardiac output states, or when pulmonary and systemic vascular resistances are similar ($PVR \approx SVR$), the pressure difference between the aorta and pulmonary artery may be insufficient to accelerate flow, resulting in low measured velocities despite a restrictive ductal caliber. Therefore, Doppler velocities across the PDA must always be interpreted within the broader hemodynamic context, including systemic blood pressure, ventricular outputs estimate (or at least stroke distance), estimated RV systolic pressure, and both right and left ventricular size and function. In this framework, velocity reflects not only ductal restriction but also the instantaneous pressure gradient across the ductus. Clinically, a "closing" Doppler pattern—defined by continuous high-velocity flow with minimal pulsatility between systole and diastole—is a hallmark of a constricted and restrictive ductus with a significant pressure difference between the 2 circulatory compartments. Anatomically, this restriction frequently manifests as a localized narrowing or area of constriction, which most commonly initiates at the pulmonary arterial end of the vessel. While a wide ductal ampulla may be visible at the aortic end, the effective restriction is determined by the narrowest point in the channel. In the context of certain congenital heart defects, such as pulmonary atresia, the ductus may be inherently small and restrictive from birth due to reduced flow in utero (decreased flow = decreased growth). Furthermore, in premature infants, a high velocity through a small duct suggests that while the shunt remains present, the vessel is successfully constricting and narrowing its lumen. Of note, even when a ductus appears restrictive, it can still carry a meaningful amount of left-to-right flow when PVR is substantially lower than SVR, creating a large pressure gradient. Therefore, ductal size or Doppler velocity alone should not be used to determine

hemodynamic significance. A comprehensive assessment must integrate additional echocardiographic markers, particularly evidence of volume loading such as left atrial and left ventricular dilation, along with other indices of significance discussed elsewhere. Importantly, this interpretation may be altered by associated lesions. In the presence of a large atrial septal defect (ASD), left-sided dilation may be attenuated or absent despite significant shunting, as the left atrium can decompress into the right atrium and right ventricle. In this situation, the volume load is redistributed to the right heart. However, in neonates, the right ventricle may accommodate increased volume with limited early dilation due to its relative compliance, which can further complicate the assessment of shunt significance. Also, the intrinsic resistance in a vessel like the ductus arteriosus is directly related to viscosity, as defined by the [Poiseuille equation](#). According to this physical law, the resistance (R) to blood flow is determined by the formula $R = 8\eta l / (\pi r^4)$, where η (eta) represents the viscosity of the fluid, l is the length of the tube, and r is its radius. Blood viscosity is largely determined by hematocrit levels. While viscosity remains relatively stable when the hematocrit is between 35% and 45%, it increases dramatically when the hematocrit rises above 50%, and even more steeply beyond 60% to 65%. Viscous energy loss is a factor that can lead to an underestimation of pressure gradients in small, tortuous, or tunnel-like PDAs (e.g., those with a diameter less than 3 mm and length greater than 10 mm). While the major determinants of PDA shunt volume are often listed as the diameter and length of the ductus, viscosity is an essential physical component of the "intrinsic ductal resistance" that limits flow in restrictive ducts. In these cases, the "path of least resistance" that blood follows is determined not just by the vessel's size, but by the properties of the blood flowing through it. More information in the [Understanding PDA Spectral Doppler](#) section.

Background

The definition of "restriction" in the context of patent ductus arteriosus (PDA) remains somewhat ambiguous, as no universally accepted criteria exist. However, it is generally understood that a "restrictive" structure limits the transmission of flow or pressure. This concept can be simplified as follows:

- Unrestrictive PDA: This allows pressure to equalize across both sides of the connected structure, similar to two boxes connected by a large channel.
- Restrictive PDA: This limits the amount of pressure or flow, akin to nearly closing off a garden hose.
- The "Most" Restrictive PDA is theoretically the one that is closed. Hence, there is no flow through it and no pressure transmission across it. Sometimes, the duct is on the verge of closing, where minimal or no blood flow is observed, especially during diastole when pressure in the aorta and pulmonary artery decreases. During systole, there may be some "stenting" of the lumen of the duct with some flow, which closes off in Diastole. We often call this "closing pattern".

Article <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7717963/> mentions: "Non-restrictive shunts have a low peak systolic velocity with a high systolic to end-diastolic velocity gradient while restrictive shunts have a high peak systolic velocity and a low systolic to diastolic velocity gradient. **If the ratio between peak systolic and end-diastolic velocity is >2 then it is considered as a pulsatile flow pattern while a ratio of <2 is described as restrictive shunt suggestive of a closing PDA**". A typical Doppler pattern for a restrictive (medium or small) PDA shows accelerated flow (e.g., 3-4 m/s, sometimes up to 6 m/s) with a diastolic prolongation, indicating continuous shunting in both systole and diastole.

Important caveats:

- "High-Velocity Flow": A restrictive PDA typically shows high-velocity flow due to the narrowed ductal lumen. Doppler studies often demonstrate peak systolic velocities greater than 2.5 m/s and sometimes exceeding 3.5 m/s.
- If velocity is less than 2 m/s - PW-Doppler can be used. Beyond 2 m/s it is better to use CW-Doppler to capture the peak velocities.
- "Continuous Flow Pattern": The flow across a restrictive PDA is typically continuous throughout the cardiac cycle, with a high-velocity systolic jet and less pronounced diastolic flow. A restrictive PDA often presents with a significant pressure gradient between the pulmonary artery and the aorta, which can be calculated using the peak Doppler velocity (using the modified Bernoulli equation). A high gradient (e.g., >20 mmHg) suggests a restrictive flow.
- The velocity criteria depend on the pressure (and/or resistance) differences between the aorta (AO) and the pulmonary artery (PA). For example, you can have a very small duct that restricts red blood cells from passing through, but if the AO and PA pressures are nearly equal, the flow will be of low velocity. Consider a situation where systolic blood pressure (sBP) is 100 mmHg and systolic pulmonary arterial pressure (sPAP) is 98 mmHg. In this case, very little flow will pass through the ductus (even if it is very tiny), but since sBP is slightly higher than sPAP, the flow will still be left-to-right. However, with only a 2 mmHg difference between the two arteries, the flow will be minimal. If the PDA is extremely narrow, such as (hypothetically) 0.001 mm, it becomes "restrictive" in that it doesn't transmit flow or pressure effectively, but the flow remains of very low velocity. Typically, at this stage, the flow contour would be continuous and non-pulsatile due to the [Windkessel Effect](#).
 - The gradient between PA and AO also depends on their respective pressures. For instance, consider a PDA connecting an aorta with pressures of 50/30 mmHg and a pulmonary artery with pressures of 20/10 mmHg. The pressure gradient across the PDA would be a peak of 30 mmHg and an end value of 20 mmHg. The flow would appear continuous and non-pulsatile, yet with a relatively high velocity, such as during systole, where the velocity might reach 2.7 m/s ($30 \text{ mmHg} \sim 4V^2$).
 - Let's consider an unrestrictive scenario: in this case, all pressure is transmitted across both ends, and the flow will depend on the pulmonary vascular resistance (PVR) to systemic vascular resistance (SVR) ratio. If the aortic pressure is 45/25 mmHg and the pulmonary artery pressure is also 45/25 mmHg, the flow will be from left to right. Unlike the previous example of 50/30 mmHg, this pressure is slightly lower, likely due to the loss of a few mmHg as the flow needs to fill a much larger vascular network, with both the pulmonary and systemic systems being connected.

In life, very few things are absolute. The PDA is a dynamic structure that gradually becomes more restrictive as it closes. As it narrows, it increasingly limits pressure transmission, much like a hose that becomes progressively occluded, restricting the flow.

Some Criteria for a restrictive ductus:

- Peak Doppler velocity > 2 m/s
- Ductal diameter < 1.5 mm
- High pressure gradient across the ductus (>20-30 mmHg)
- Continuous Doppler flow with a "sawtooth" appearance on spectral Doppler.

paper: <https://www.ahajournals.org/doi/full/10.1161/CIRCULATIONAHA.105.592063> says "If the ductus is large and offers minimal resistance to flow (nonrestrictive), the degree of shunting depends on the status of the pulmonary vascular resistance. In many children with moderate or large patent ductus, pulmonary vascular resistance remains modestly elevated, which limits the shunting sufficiently to lessen the physiological impact and permit survival and growth"

This paper <https://jamanetwork.com/journals/jamapediatrics/article-abstract/2375129> says: "For a PDA to cause significant shunting from systemic to pulmonary circulation, the flow needs to be unrestrictive and completely or almost completely left to right; the latter is sometimes referred to as a growing pattern. The PDAs of less than 1.5 mm in diameter are considered small because they are commonly restrictive, cause a mild increase in pulmonary circulation, and are rarely associated with echocardiographic markers of a high-volume shunt.^{12,13} Further subclassification of PDAs of at least 1.5 mm as moderate or large is based on incremental probability of a high-volume shunt. Although the diameter correlates well with shunt volume, using an absolute cutoff to describe hemodynamic significance may be misleading given the interobserver measurement variability (10%-15%) and its dynamic nature."

This paper <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1720677/> says: "The closing pattern suggests a restrictive PDA small enough to prevent a clinically significant left to right shunt. The growing and pulsatile patterns reflect falling pulmonary artery resistance in the presence of a large ductus with clinically significant left to right shunt, and are an indicator for risk of developing clinically significant PDA. Clinicians need to recognise that once pulmonary artery resistance has fallen, if the closing pattern has not developed, the ductus is large enough to create problems."

